**1** **(a)** 11 **[B1]**

**(b)** 61 **[B2]**

M1 for 64 or 3 seen.

*[Total Marks: 3]*

**2** **(a)** $8 \times 9$ **[M1]**

72 **[A1]**

**(b)** 5 **[B1]**

*[Total Marks: 3]*

**3** **(a)** 21 **[B2]**

M1 for 13 or 8

**(b)** 80 **[B2]**

M1 for 16 or 5

*[Total Marks: 4]*

**4** **(a)** Calculates $3^3 = 27$ and $\sqrt{144} = 12$ **[M1]**

15 **[A1]**

**(b)** Calculates $2^4 = 16$ and $\sqrt[3]{125} = 5$ **[M1]**

80 **[A1]**

*[Total Marks: 4]*

**5** **(a)** 14 **[B1]**

**(b)** 59 **[B2]**

If 0 scored, M1 for 64 or 5 seen **[M1]**

**(c)** 5 **[B2]**

If 0 scored, M1 for $169 - 144$ or 25 seen **[M1]**

*[Total Marks: 5]*

**6** **(a)** 14 **[B1]**

**(b)** 48 **[B2]**

If 0 scored, allow [M1] for correctly evaluating either $\sqrt[3]{27} = 3$ or $2^4 = 16$

**(c)** 8.8 or $\frac{44}{5}$ **[B2]**

If 0 scored, allow [M1] for a correct numerator of 44 or a correct denominator of 5

*[Total Marks: 5]*

**7** **(a)** $\frac{37}{7}$ or $5\frac{2}{7}$ **[B2]**

M1 for $25 + 12$ or $8 - 1$ seen.

**(b)** 4 **[B2]**

M1 for $9 \times 9$ or $3^2 \times 3^2$ or 81 seen.

*[Total Marks: 4]*

**8** **(a)** Numerator correctly evaluated to $64 + 15 = 79$ **[B1]**

Denominator correctly evaluated to 4 **[B1]**

19.75 **[B1]**

**(b)** Substitution leading to $25 - (-8)$ **[M1]**

33 **[A1]**

*[Total Marks: 5]*

**9** **(a)** 5 **[B1]**

**(b)** 150 **[B2]**

If 0 scored, allow [M1] for $6 \times 5^2$ or $6 \times 25$

**(c)** 121 **[B2]**

If 0 scored, allow [M1] for evaluating $3^3 - 4^2 = 11$

*[Total Marks: 5]*

**10** **(a)** $\frac{15 + 10}{25}$ **[M1]**

$\left(\frac{25}{25}\right)^{-2}$ or $1^{-2}$ **[M1]**

1 **[A1]**

**(b)** $x^{\frac{2}{3}} = 4$ **[M1]**

$x = 4^{\frac{3}{2}}$ or $(\sqrt{4})^3$ or $x^2 = 4^3 = 64$ **[M1]**

8 **[A1]**

*[Total Marks: 6]*

**11** **(a)** $3x + 4 = 64$ **[M1]**

$3x = 60$ **[M1]**

20 **[A1]**

**(b)** $x^2 = 64 + 36 = 100$ **[M1]**

$10, -10$ **[A1]**

**(c)** $3\sqrt{x} = 21$ or $\sqrt{x} = 7$ **[M1]**

49 **[A1]**

*[Total Marks: 7]*

**12 (a)** 125 **[B2]**

If 0 scored, M1 for evaluating $16 - 9 - 2$ to get 5 **[M1]**

**(b)** $-2$ **[B3]**

M1 for $\sqrt{64}$ or 8 **[M1]**

M1 for $5 - 9$ or $-4$ **[M1]**

**(c)** 729 **[B3]**

M1 for $2\sqrt[3]{y} = 6$ or $\sqrt[3]{y} = 3$ **[M1]**

M1 for $y = 27$ **[M1]**

*[Total Marks: 8]*

**13 (a)** 5.3990299... **[B2]**

M1 for numerator evaluated to 163.91... or denominator evaluated to 30.359...

**(b)** 5.399 **[B1]**

**(c)** 6 **[B2]**

M1 for evaluating $6^4 = 1296$ and $7^4 = 2401$

*[Total Marks: 5]*

**14 (a) (i)** $1.49315\ldots$ **[B2]**

If 0 scored, M1 for 3.329 seen or 72.25 and 68.921 **[M1]**

**(ii)** 1.49 **[B1]**

**(b)** 96 **[B2]**

If 0 scored, M1 for 23.04 or 0.24 seen **[M1]**

*[Total Marks: 5]*

**15 (a)** $\sqrt{151.83}$ or $12.32\ldots$ or 29.791 **[M1]**

$0.4136\ldots$ **[A1]**

**(b)** $p \times q = 3402$ **[M1]**

$\sqrt[3]{3402}$ or $15.0401\ldots$ **[M1]**

15.0 **[A1]**

*[Total Marks: 5]*

**16** **(a)** 1.59 **[B2]**

M1 for $23.02\ldots$ or 14.44 seen.

**(b)** 1734 **[B3]**

M1 for $\sqrt[3]{4913}$ or 17 seen.

M1 for $6 \times 17^2$.

*[Total Marks: 5]*

**17** **(a)** 1.5 **[B1]**

**(b)** Rearranges to $2x^4 = 1250$ leading to $x^4 = 625$ **[M1]**

$x = 5$ **[B1]**

$x = -5$ **[B1]**

**(c)** Correct substitution: $\sqrt{\dfrac{3(4.2)^2 + (-5.7)}{1.8}}$ **[M1]**

Evaluates numerator to 47.22 OR fraction to $26.2333\ldots$ **[M1]**

5.12 **[A1]**

*[Total Marks: 7]*

**18** **(a)** 0.239 **[B2]**

If 0 scored, allow [M1] for evaluating the numerator as 19.679... or the denominator as 82.464...

**(b)** 64 **[B3]**

If 0 scored, allow [M1] for isolating $x^2$ correctly, e.g., $x^2 = 16^3$ or $x^2 = 4096$

Allow a further [M1] for $x = \sqrt{4096}$

**(c)** 7.44 or 7.443... **[B3]**

If 0 scored, allow [M1] for finding the volume of the cube as $12^3$ or 1728

Allow a further [M1] for correctly rearranging the sphere volume formula to solve for $r$, e.g., $r^3 = \dfrac{1728 \times 3}{4\pi}$ or $r = \sqrt[3]{\dfrac{1296}{\pi}}$

*[Total Marks: 8]*